

AMENDMENT NO. 2 JULY 2024

TO

IS 15546 : 2005 AUTOMOTIVE VEHICLES — SEATS, THEIR ANCHORAGES AND HEAD RESTRAINTS FOR CATEGORY M1 — SPECIFICATION

(Page 2, clause 3.15) — Substitute the following for the existing:

‘3.15 Partitioning System — It is the parts or devices which, in addition to the seat-backs, that is intended to protect the occupants from displaced luggage, in particular, a partitioning system maybe constituted by netting or wire mesh located above the level of the seat backs in their upright or folded down position. Head restraints fitted as standard equipment for vehicles equipped with such parts or devices shall be considered as part of the partitioning system. However, a seat equipped with a head restraint shall not be considered as being on its own a partitioning system.’

(Page 5, clause 4.8.1) — Substitute the following for the existing:

‘4.8.1 Seat-Backs

Seat-backs and/or head restraints located such that they constitute the forward boundary of the luggage compartment, all seats being in place and in the normal position of use as indicated by the manufacturer, shall have sufficient strength to protect the occupants from displaced luggage in a frontal impact. This requirement is deemed to be met if, during and after the test described in 5.9, the seat-backs remain in position and the locking mechanisms remain in place. However, the deformation of the seat-backs and their fastenings during the test is permitted, provided that the forward contour of the tested seat-back and/or head restraint, that are harder than 50 shore. A does not move forward of a transverse vertical plane which passes through

- a) A point of 150 mm forward of the R point of the seat in question, for the parts of the head restraint; and
- b) A point of 100 mm forward of the R point of the seat in question, for parts of the seat-back; excluding the rebound phases of the test blocks.

For integrated head restraints, the limit between the head restraint and the seat-back is defined by the plane perpendicular to the reference line 540 mm from the R point.

All measurements shall be taken in the longitudinal median plane of the corresponding seat or seating position for each seating position constituting the forward boundary of the luggage compartment. During the test described in 5.9, the test blocks shall remain behind the seat-back(s) in question.

In case of damage to a safety-belt retractor, it shall be verified that the retractor is locked as a result of the test or that it can be locked by a manual pull out of the strap.’

(Page 6, clause 4.8.2.2) — Substitute the following for the existing:

‘4.8.2.2 This requirement is deemed to be met if, during the test, the partitioning systems remain in position. However, the deformation of the partitioning systems during the test is permitted, provided that the forward contour of the parts of the partitioning systems [(including parts of the tested seat-back(s) and/or head restraint(s)] that are harder than 50 shore A, does not move forward of transverse vertical plane which passes through a point of 150 mm forward of R point of the seats in question, for the parts of the head restraint and a point of 100 mm forward of R point of the seats in question, for the parts of seat-back excluding the rebound phases of the test blocks. All measurements shall be taken in the longitudinal median plane of the corresponding seat or seating position for each seating position constituting the forward boundary of the luggage compartment.’

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(Page 6, clause **4.8.2.3**) — Substitute the following for the existing:

‘4.8.2.3 After the test, no sharp or rough edges likely to increase the danger or severity of injuries of the occupants shall be present.

In case of damage to a safety-belt retractor, it shall be verified that the retractor is locked already or that it can be locked by a manual pull out of the strap.’

(Page 6, clause **4.8.3**) — Delete.

(Page 6, clause **4.8.3**) — Renumber **4.8.4** as **4.8.3**

‘4.8.3 The requirements mentioned in **4.8.1** and **4.8.2** shall not apply to luggage retention systems which are activated automatically in case of an impact. The manufacturer shall demonstrate to the satisfaction of the test agency that the protection offered by such systems is equivalent to that described in **4.8.1** and **4.8.2.**’

(Page 9, clause **5.4.2.8**) — Substitute the following for existing:

‘5.4.2.8 To check the strength of the head restraint, the initial load specified in **5.4.2.3** and **5.4.2.5** are increased to 890 N unless the breakage of the seat or seat-back occurs earlier.’

(Page 11, clause **5.9.1**) — Substitute the following for existing:

‘5.9.1 Test Blocks

Rigid blocks, with the centre of inertia in the geometric centre.

Type 1

Dimensions : 300 mm x 300 mm x 300 mm with all edges and comers rounded to 20 mm.

Mass :18 kg

Moment of inertia 0.3 +/- 0.05 kgm² (around all 3 principal axis of inertia of the luggage blocks)

Type 2

Dimensions : 500 mm x 350 mm x 125 mm with all edges and corners rounded to 20 mm.

Mass : 10 kg’

(Page 11, clause **5.9.3.1**) — Substitute the following for the existing:

‘5.9.3.1 General requirements

At the option of the car manufacturer, parts whose hardness is lower than 50 shore A can be removed from the tested seat and head restraint for the tests.

Two numbers of Type 1 test blocks shall be placed on the floor of the luggage compartment. In order to determine the location of the test blocks in the longitudinal direction, they shall first be positioned such that their front side contacts that part of the vehicle which constitutes the forward boundary of the luggage compartment and that their lower side rests on the floor of the luggage compartment. They shall then be moved backwards and parallel to the longitudinal median plane of the vehicle until their geometrical centre has transverse a horizontal distance of 200 mm. if the dimensions of the luggage compartment do not allow a distance of 200 mm and if the rear seats are horizontally adjustable, these seats shall be moved forward to the limit of the adjustment range intended for normal occupant use, or to the position resulting in a distance of 200 mm, whichever is less. In other cases, the test blocks shall be placed as far as possible behind the rear seats. The distance between the longitudinal median plane of the vehicle and the inward facing side of each test block shall be 25 mm to obtain a distance of 50 mm between both blocks.

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During the test, the seats must be adjusted to ensure that the locking system cannot be released by external factors, if applicable, the seats shall be adjusted as follows:

The longitudinal adjustment shall be secured one notch or 10 mm in front of the rearmost possible position of use specified by the manufacturer (for seats with independent vertical adjustment, the cushion shall be placed to its lowest possible position). The test shall be carried out with the seat-backs in their normal position of use.

If the seat-back is fitted with a head restraint, the test must be carried out with the head restraint placed in the highest position, if adjustable.

If the back(s) of the rear seat(s) can be folded down, they shall be secured in their upright normal position by the standard locking mechanism.

Seats behind which the Type 1 blocks cannot be installed are exempted from this test.

All seating positions of the seat row under test shall be fitted with all components of its safety-belt providing the restraining function that are part of the seat.'

(Page 13, clause 5.9.4) — Substitute the following for existing:

‘5.9.4 Dynamic Testing Seat-Backs and Partitioning Systems Used as Luggage Restraint Systems

The body of the passenger car shall be anchored securely to a test sled, and this anchorage shall not act as reinforcement for seat-backs and the partitioning system. After the installation of the test blocks as described in 5.9.3.1 or 5.9.3.2, the passenger car body shall be decelerated or, at the choice of the applicant, accelerated such that the curve remains within the area of the graph as shown in the Fig. 4, and the total velocity change ΔV is 50 ± 2 km/h. With the agreement of the manufacturer, the above-described test pulse corridor can be used alternatively to fulfil the test of the seat strength according to 5.3.1.

However, if the tests were performed at a higher speed and/or the acceleration/deceleration curve have exceeded the upper level of the hatched area and the seats meet the requirements, the test shall be considered satisfactory based on the mutual understanding between Vehicle Manufacturer and Test Agency.'

(Page 13, Fig 4) — Substitute the following for existing:

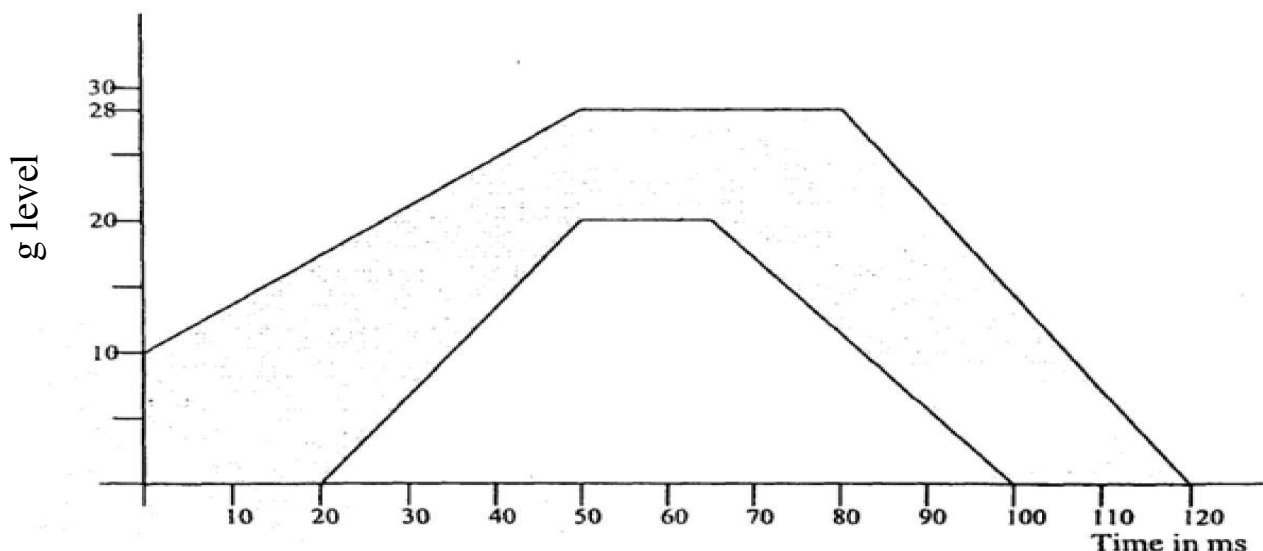


FIG. 4 CORRIDOR OF SLED'S DECELERATION OR ACCELERATION AS A FUNCTION OF TIME (SIMULATION OF FRONTAL IMPACT)'